Quantification: I534_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	13
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2941.54	2951	1607.61	px^2
major_axis_length_px	lacuna shape	85.8598	83.1379	30.1528	px
minor_axis_length_px	lacuna shape	45.6621	45.1555	14.9313	px
aspect_ratio	lacuna shape	1.9335	1.9125	0.5477	unitless
eccentricity	lacuna shape	0.8143	0.8524	0.1082	unitless
solidity	lacuna shape	0.8338	0.8509	0.098	unitless
roots_count	headline	7.3077	6	4.9055	count
ring_length_r30_px	headline	252.692	231	129.085	px
ring_length_r60_px	reported	572.462	533	211.409	px
owned_length_px	ownership-dependent	712.713	585.73	552.206	px
edge_count	ownership-dependent	33.7692	24	26.4926	count
mean_edge_length_px	ownership-dependent	21.4108	21.2013	4.2347	px
perimeter_px	normalised	273.55	262.877	100.444	px
ring_area_r30_px2	normalised	9806.31	9533	2617.99	px^2
ring_area_r60_px2	normalised	22647.7	22663	5117.86	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.99	1	0.0267	unitless
ring_density_r30	normalised	0.024848	0.0231267	0.00843234	px^-1
ring_density_r60	normalised	0.0250018	0.02222	0.00679102	px^-1
roots_per_100px_perimeter	normalised	2.4771	2.5172	0.8907	per 100 px
ring_attached_length_r30_px	network v2	214.615	198	130.226	px
ring_attached_length_r60_px	network v2	390.385	373	223.194	px
ring_length_w_r30_px	network v2	287.407	258.309	147.665	px
ring_length_w_r60_px	network v2	652.42	602.973	239.185	px
sholl_crossings_r10	network v2	5.8462	6	2.8239	count
sholl_crossings_r20	network v2	7.3846	6	4.0935	count
sholl_crossings_r30	network v2	8.2308	8	3.5392	count
loop_count	loops, report only	1.6923	1	2.4962	count
loop_edge_share	loops, report only	0.128	0.0678	0.145	share
ring_width_mean_r30_px	width	7.7624	7.6554	0.5221	px

Per field
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measure	value	units
analysed_area_px2	877616	px^2
total_skeleton_length_px	22729	px
canalicular_length_density_per_px	0.0258986	px^-1
junction_count	432	count
junction_density_per_px2	0.00049224	px^-2
lacuna_count	17	count

lacunae_per_px2	1.937e-05	px^-2
skeleton_component_count	311	count
mean_component_length_px	73.0836	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0294043	px^-1
field_density_without_flagged_per_px	0.0228594	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	225	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0258986	px^-1
field_length_density_with_unattached_per_px	0.0261549	px^-1
excluded_area_px2	127816	px^2
analysed_field_area_px2	920760	px^2
loop_count	36	count
loop_edge_share	0.0975	share
edge_count	1220	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8449	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	22455	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	718	47.2395	23.002	2.0537	0.8734	0.8022	-1.0816	93.7911	437.035	False
2	2720	92.088	38.5852	2.3866	0.908	0.9547	-1.1428	215.941	986.257	True
3	6078	103.767	76.7807	1.3515	0.6727	0.9519	-0.9596	258.575	697.738	False
4	749	51.0208	20.659	2.4697	0.9144	0.8874	1.1423	366.493	18.0734	True
5	2951	92.7411	57.1766	1.622	0.7873	0.6519	0.6626	412.099	797.415	False
6	3683	112.658	46.4293	2.4264	0.9111	0.7914	-0.9592	433.344	623.77	False
7	623	33.6165	26.9149	1.249	0.5991	0.8534	-1.4105	448.144	1010.89	True
8	3483	74.1218	60.6446	1.2222	0.575	0.9286	-0.204	483.984	66.1174	False
9	3784	85.9898	57.0703	1.5067	0.748	0.9309	-0.1766	493.888	417.965	False
10	4829	127.96	50.7336	2.5222	0.918	0.7542	-1.2747	583.146	228.783	False
11	2220	83.1379	43.4719	1.9125	0.8524	0.7754	0.0297	577.841	853.017	False
12	4318	144.296	45.1555	3.1955	0.9498	0.6869	1.4075	580.847	622.689	False
13	2037	75.2328	35.6451	2.1106	0.8806	0.9122	-0.2022	721.922	650.957	False
14	1103	54.1002	27.7589	1.9489	0.8583	0.9026	-0.7329	709.348	70.3019	False
15	812	45.0748	28.6241	1.5747	0.7725	0.7785	0.6665	738.431	13.2463	True
16	1992	69.0062	38.1155	1.8104	0.8336	0.9009	-1.2002	817.186	92.8996	False
17	1044	45.928	31.6235	1.4523	0.7252	0.8509	0.2821	817.663	289.875	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	2	231	643	1631.07	87	18.7479	8.2543
2	3	55	210	61.7696	5	12.3539	6.4488
3	6	275	685	261.309	13	20.1007	7.3739
4	3	126	307	183.61	6	30.6017	6.8573
5	13	460	1038	1639.39	77	21.2908	7.061
6	11	312	500	304.836	16	19.0522	7.5715
7	4	130	342	352.049	14	25.1463	7.5961
8	5	172	447	347.309	19	18.2794	8.499
9	4	157	533	184.681	12	15.3901	7.7554
10	18	496	814	820.411	50	16.4082	7.4463
11	7	270	645	842.909	34	24.7914	7.6554
12	13	376	691	683.6	31	22.0516	8.0545
13	3	147	421	1516.09	59	25.6965	8.8704
14	4	93	231	169.61	8	21.2013	7.4866
15	4	68	135	49.5563	5	9.9113	5.5824
16	6	190	379	585.73	24	24.4054	7.2164
17	3	106	415	278.321	9	30.9245	7.6667

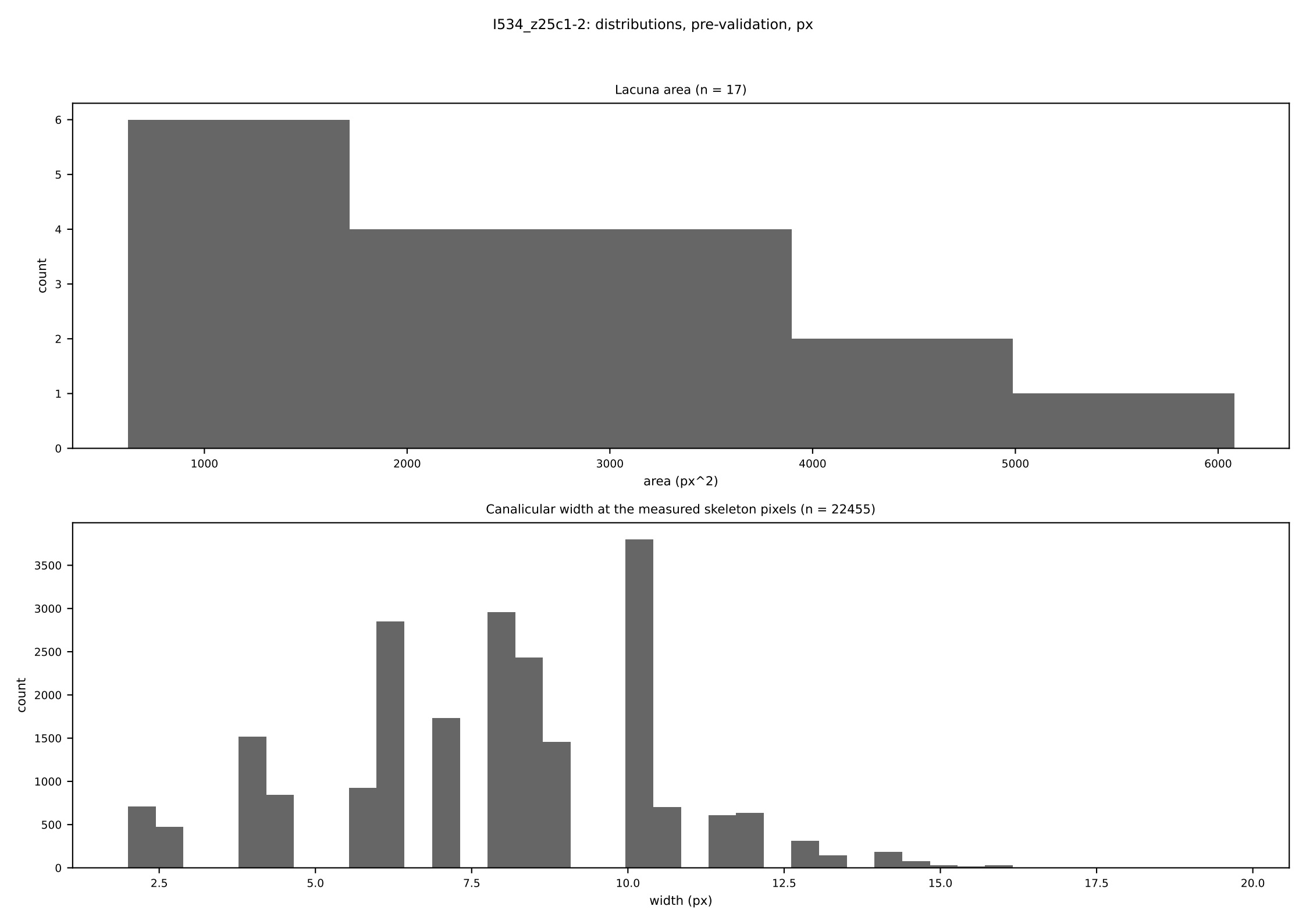
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	132.125	6322	18207	1	1	0.0365391	0.0353161	1.5137
2	237.965	7174	17084	0.7631	0.6986	0.00766657	0.0122922	1.2607
3	334.475	11891	27068	1	1	0.0231267	0.0253066	1.7939
4	128.947	4467	9758	0.694	0.6341	0.0282068	0.0314614	2.3265
5	419.345	11919	24709	1	1	0.0385938	0.042009	3.1001
6	332.676	11820	24257	1	1	0.0263959	0.0206126	3.3065
7	114.184	3796	10383	0.6248	0.5844	0.0342466	0.0329385	3.5031
8	249.078	9533	20117	1	0.9067	0.0180426	0.02222	2.0074
9	262.877	9914	25300	1	1	0.0158362	0.0210672	1.5216
10	405.588	13364	31075	1	1	0.0371146	0.0261947	4.438
11	278.09	9508	22663	1	1	0.0283971	0.0284605	2.5172
12	406.416	13768	29555	1	1	0.0273097	0.0233801	3.1987
13	220.35	8451	20862	1	1	0.0173944	0.0201802	1.3615
14	153.238	5825	12202	1	0.9659	0.0159657	0.0189313	2.6103
15	188.368	3275	4996	0.6244	0.5829	0.0207634	0.0270216	2.1235
16	215.037	8340	19166	1	0.9972	0.0227818	0.0197746	2.7902
17	146.853	6827	19239	1	1	0.0155266	0.0215708	2.0429

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	198	514	258.309	737.512	7	5	9
2	55	55	66.1838	236.995	2	1	0
3	224	373	300.238	767.541	7	6	8
4	114	152	139.74	352.806	3	2	6
5	416	877	530.345	1180.76	10	16	16
6	295	448	364.262	573.215	6	9	9
7	109	303	145.397	390.191	2	3	2
8	125	276	191.196	502.961	3	7	6
9	69	130	173.468	602.973	2	4	6
10	480	694	566.801	924.879	11	15	9
11	211	456	305.764	751.009	6	9	11
12	314	491	424.363	777.612	8	8	13
13	97	179	169.196	489.073	4	6	6
14	76	125	111.225	266.764	2	4	5
15	52	52	70.0416	151.711	1	0	3

16	182	297	215.338	424.363	6	4	6
17	103	215	125.782	482.801	4	3	3


I534_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 17)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22455)
count
width (px)